

**FILING OF PETITION FOR
MULTI YEAR TARIFF
FOR CONTROL PERIOD FROM FY 2016-17 TO FY 2020-21**

To



JHARKHAND STATE ELECTRICITY REGULATORY COMMISSION (JSERC)

Submitted by:



**Tenughat Vidyut Nigam Limited (TVNL)
(A Government of Jharkhand Undertaking)
Hinoo, Doranda, Ranchi-834 002**

April, 2017



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**BEFORE THE JHARKHAND STATE ELECTRICITY REGULATORY
COMMISSION**

Filing No: _____

Case No: _____

IN THE MATTER OF:

Filing of Petition for Multi Year Tariff for Control Period of FY 2016-17 to 2020-21 for TVNL under JSERC (Terms and Conditions for Determination of Generation Tariff) Regulations, 2015 issued by the Honourable Jharkhand State Electricity Regulatory Commission (JSERC), hereinafter referred to as Honourable Commission.

AND

IN THE MATTER OF:

Tenughat Vidyut Nigam Limited ("TVNL")

Hinoo, Doranda, Ranchi

Jharkhand – 834 002

.....PETITIONER

The Petitioner respectfully submits



CHAPTER 1. INTRODUCTION

1.1 Introduction

1.1.1 Tenughat Vidyut Nigam Limited is a power generating company incorporated on 26th Nov 1987 under Indian Company's Act, 1956. With the creation of Jharkhand State on 15th November 2000 from the erstwhile Bihar State, the TVNL became an undertaking of Government of Jharkhand. The TVNL owns and operates an installed power generation capacity of 420 MW with two units of 210 MW each at Tenughat Thermal Power Station located at Lalpania at the banks of Tenughat reservoir in Bokaro district with its head office at Ranchi. The units were under commercial operation as detailed below:

(a) Unit I - September 1996

(b) Unit II - September 1997

1.1.2 TVNL further aims to consolidate its position as the leading thermal power generation company in Jharkhand and to diversify across the power value chain in Jharkhand by considering backward and forward integration into areas such as transmission, distribution, coal mining etc.

1.1.3 Petitioner has over the years made efforts to improve its performance across all the parameters in order to provide reliable and affordable supply. In this endeavour; timely receipt of its energy charges from its client is very important to ensure compliance of good O&M practices and plant betterment plans. As at 31st March 2017 JUVNL/JBVNL owes around INR 3283 Cr.

1.1.4 Petitioner would like to request the Hon'ble Commission to consider this parameter while approving the tariff as late realization of payment affects execution of its O&M Plans which in turn has inherent cost of carry and increases all the associated costs of generation.

1.2 Need for Petition

1.2.1 The Hon'ble Commission has notified the JSERC (Terms and Conditions for determination of Generation Tariff) Regulations, 2015 published on 20th Jan 2016 (Hereinafter referred to as JSERC Generation Tariff Regulations, 2015) in exercise of the powers conferred by Section 61 and 62 read with Section 181 of the Electricity Act, 2003. The sub-clause (3) of Clause 5 of these regulations require the generating company, to prepare and file MYT projections for Control Period i.e. FY 2016-17 to FY 2020-21 for getting the same approved by the Honourable Commission. In compliance of this provision, the Petitioner is filing this petition.



CHAPTER 2. EXECUTIVE SUMMARY

2.1 Introduction

- 2.1.1 TVNL submits that late payment by JUVNL/JBVNL and its associated cost of carry has been uncontrollable on their part and thus any gains the company may receive by third party sale should not be shared with JUVNL/JBVNL with any rebate in regulated tariff's determination and that TVNL may be allowed to operate as an independent commercial entity.
- 2.1.2 TVNL also submits that non-payment has been impacting its financial performance; which in turn has been severely impacting its operating performance as its performance improvement plans are getting hampered. As at 31st March 2017 JUVNL/JBVNL owes around INR 3283 Cr.

2.2 Revenue Requirements for FY 2016-17 to FY 2020-21

- 2.2.1 Sections below highlight the energy charges and fixed costs for the Control period.
- 2.2.2 **Capital Investment Plan and Capitalization**
- 2.2.3 Following tables summarizes the Capital Investment Plan and Capitalization for MYT control period for FY 2016-17 to FY 2020-21:

Table 1: Summary of Capital Investment Plan

Capital Expenditure	FY 17	FY 18	FY 19	FY 20	FY 21
Control & Instrumentation	25.88	12.00	22.15	2.55	0.35
Chemistry (OPN)	0.00	0.00	0.65	0.65	0.00
Civil Works	3.51	4.00	35.50	34.00	12.50
MGR System	0.00	0.00	5.00	5.00	10.00
Electrical Dept.(EWD) Works	35.04	24.24	23.20	12.15	11.35
Turbine Dept. (TMD) Works	33.20	11.84	18.08	27.46	50.66
Boiler Dept. (BMD) Works	2.10	17.27	13.82	9.12	13.32
Others	0.69	0.00	0.00	0.00	0.00
Coal Handling Plant (CHP)	0.00	5.00	10.00	16.20	13.00
Total	100.42	74.35	128.40	107.13	111.18



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Table 2: Summary of Capitalization

Capital Expenditure	FY 17	FY 18	FY 19	FY 20	FY 21
Opening CWIP	55.24	15.57	8.99	13.74	12.09
Capex added during the year	100.42	74.35	128.40	107.13	111.18
Capex Converted to GFA	140.10	80.92	123.65	108.78	110.94
Closing CWIP	15.57	8.99	13.74	12.09	12.33
Opening GFA	1,588.50	1,728.59	1,809.52	1,933.17	2,041.95
GFA Conversion from Capitalization (90% of CWIP)	140.10	80.92	123.65	108.78	110.94
Closing GFA	1,728.59	1,809.52	1,933.17	2,041.95	2,152.89
Debt (70%)	98.07	56.65	86.56	76.15	77.66
Equity (30%)	42.03	24.28	37.10	32.63	33.28

2.2.4 Operating Parameters

2.2.5 Following table summarizes the operating parameters projected for MYT control period for FY 2016-17 to FY 2020-21:

Table 3: Summary of Operating Parameters for Second Control Period

Particular	FY 17	FY 18	FY 19	FY 20	FY 21
PAF (%)	85%	85%	85%	85%	85%
PLF (%)	85%	85%	85%	85%	85%
Station Heat Rate (Kcal/kWh)	3,043	2,908	2,773	2,638	2,503
Aux Cons (%)	9.50%	9.50%	9.50%	9.50%	9.50%
Secondary Oil Cons (ml/kWh)	1.00	1.00	1.00	1.00	1.00
Transit Loss (%)	0.20%	0.20%	0.20%	0.20%	0.20%

2.2.6 Gross and Net Generation

Following table summarizes the energy in MUs projected for MYT control period for FY 2016-17 to FY 2020-21:



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Table 4: Summary of Projected Generation for Second Control Period

Generation Details	Unit	FY 17	FY 18	FY 19	FY 20	FY 21
Installed Capacity	MW	420	420	420	420	420
Plant Load Factor	%	85%	85%	85%	85%	85%
Gross Generation	MU	3127	3127	3127	3127	3127
Auxiliary Consumption	%	9.5%	9.5%	9.5%	9.5%	9.5%
Net Generation	MU	2830	2830	2830	2830	2830

2.2.7 Energy Charges

Following table summarizes the energy charges projections for MYT control period for FY 2016-17 to FY 2020-21:

Table 5: Summary of Projected Energy Charges for Second Control Period

Computation of Energy Charges	Unit	FY 17	FY 18	FY 19	FY 20	FY 21
Gross Generation	MU	3,127	3,127	3,127	3,127	3,127
Net Generation	MU	2,830	2,830	2,830	2,830	2,830
Station Heat Rate	Kcal/kWh	3,043	2,908	2,773	2,638	2,503
Secondary Fuel Oil Consumption	ml/kWh	1	1	1	1	1
Gross Calorific Value of Coal	Kcal/Kg	3,615	3,615	3,615	3,615	3,615
Gross Calorific Value of Oil	Kcal/Kl	9,359	9,359	9,359	9,359	9,359
Total Heat Required	Kcal	9,516,435	9,094,247	8,672,058	8,249,870	7,827,682
Oil Consumption	KL	3,127	3,127	3,127	3,127	3,127
Heat From Oil	Kcal	29,269	29,269	29,269	29,269	29,269
Heat From Coal	Kcal	9,487,166	9,064,978	8,642,790	8,220,602	7,798,413
Transit Loss	%	0.20%	0.20%	0.20%	0.20%	0.20%
Total Coal Consumption including Transit Loss	MT/ year	2,629,469	2,512,455	2,395,441	2,278,427	2,161,413
Cost of Oil	Rs /KL	36,005	36,005	36,005	36,005	36,005
Cost of Coal	Rs /MT	2,458	2,458	2,458	2,458	2,458
Total Cost of Oil	Rs Cr	11.26	11.26	11.26	11.26	11.26
Total Cost of Coal	Rs Cr	646.35	617.59	588.83	560.06	531.30
Energy Charges (Rs/kWh)	Rs/kWh	2.32	2.22	2.12	2.02	1.92



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2.2.8 Fixed Costs

The table given below summarises total fixed cost projected for each year of the Control Period FY 2016-17 to 2020-21:

Table 6: Summary of Fixed Costs projected for Second Control Period

Fixed Cost Summary	FY 17	FY 18	FY 19	FY 20	FY 21
O&M Expenses	171.09	188.76	198.23	206.96	217.65
Depreciation	82.78	88.62	94.02	100.16	105.97
Interest on Loan	144.68	149.25	152.75	156.91	160.09
Return on Equity	50.07	55.21	59.96	65.37	70.48
Interest on Working Capital	40.64	40.92	40.69	40.45	40.30
Water Charges	12.00	12.00	12.00	12.00	12.00
Less: Non-Tariff Income	(34.29)	(34.29)	(34.29)	(34.29)	(34.29)
Total fixed Cost	466.97	500.47	523.37	547.56	572.19

2.2.9 Total costs to be covered

Following table summarizes the Revenue requirement computations for each year of the Control Period FY 2016-17 to FY 2020-21:

Table 7: Summary of Total Costs projected for Second Control Period

Revenue Requirement	Unit	FY 17	FY 18	FY 19	FY 20	FY 21
Fixed Charges	Rs Cr	501.26	534.76	557.66	581.85	606.48
Energy Charge Rate	Rs/kWh	2.32	2.22	2.12	2.02	1.92
Other Income	Rs Cr	(34.29)	(34.29)	(34.29)	(34.29)	(34.29)
Fixed Income to be raised through tariff	Rs Cr	466.97	500.47	523.37	547.56	572.19



CHAPTER 3. COMPLIANCE OF DIRECTIVES

The following table summarizes the Petitioner's response to directives given by the Hon'ble Commission in the previous tariff orders:

Table 8: Status on existing directives issued by Hon'ble Commission

Directives	Compliance to earlier Directives	Views of the Commission	Current Situation
Auditor's certificate for additional capitalization and Fixed Asset Register	The petitioner submitted that it is trying hard to comply with Commissions directives. The Petitioner further submitted that the management has invited tenders from appropriate firms for the work, and will award the same to competent firms in short time. The Petitioner requested the Commission to exempt it in this regard as this exercise would take some time. Further, the Petitioner submitted that it has commissioned the MGR system and the trial run of the MGR system has been completed on 25.06.2014.	The Commission directs the Petitioner to report the status of the compliance of this directive within one month of date of issue of this Tariff Order.	The Petitioner submits to the Hon'ble Commission that the Fixed Asset Register till FY 2015-16 has been prepared and FAR for FY 2016-17 is under preparation. The Petitioner further submits that the MGR system had been capitalised in FY 2015-16 and it was taken into operation in October 2015. Also, TVNL is transporting 90% of its coal requirement from this MGR system.
SLDC Certificate for PAF	The Petitioner mentioned that it had submitted the same along with the Petition.	The same was not submitted along with the petition and was submitted later as part of the response to the discrepancy note raised by the Commission. The Commission directs the Petitioner to submit the requisite SLDC certificates along with the next Tariff Petition	The Petitioner submits to the Hon'ble Commission that it has been maintaining SLDC certificates for all months and will submit the same during true-up for respective years.



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Directives	Compliance to earlier Directives	Views of the Commission	Current Situation
Auxiliary Consumption – Action Plan to reduce	The Petitioner submitted that it had completed the process of re-calibration of meters for measuring auxiliary consumption. Further it was added that the Petitioner will install fresh energy meters at various locations to calculate area wise power consumption. The Petitioner also submitted that it has been making all possible efforts for reduction of Auxiliary Consumption, however due to constraints like ageing, partial loading, quality of coal etc. it was not being able to achieve the same to the desired value.	The Commission directs the Petitioner to submit detailed progress report on the same within one month of the date of issue of this Tariff Order.	The Petitioner submits that it has been making all possible efforts for reducing the Auxiliary Consumption, however due to constraints like ageing, partial loading, quality of coal etc. it was not being able to achieve the same to the desired value.
Secondary Fuel Oil Consumption	The Petitioner submitted that it has been making all possible efforts for reducing the secondary oil consumption and bring down the same to the specified normative range. The Petitioner further submitted that considering the operational performance of FY 2014-15, the same has been successfully come down to 1.49 (ml/kWh) as compared to 1.88(ml/kWh) in FY 2013-14. The Petitioner further hoped to successfully reduce and bring down the same within specified normative range in near future.	The Commission appreciates the efforts of the Petitioner. However it is a matter of concern that the SFC is still higher than normative levels. The Petitioner is directed to bring down Secondary Fuel Oil consumption to the normative range.	The Petitioner submits that it has been making all possible efforts for reducing the secondary oil consumption and bring down the same to the specified normative range.
Ash Utilization and Management	The Petitioner submitted that it has approached to cement manufacturers like Ultratech cement Co. & Burnpur Cement Co.	The Commission views that the Petitioner has not taken the directive of the Commission seriously and	1. Currently, TVNL has tied-up with brick manufacturer for taking around 17,000 MT/month of ash from bottom of ESP



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Directives	Compliance to earlier Directives	Views of the Commission	Current Situation
	for setting cement plant nearby Tenughat TPS. Both cement manufacturers had visited site for setting up Cement Plants in the nearby areas of Tenughat TPS. But nothing has materialized till date.	directs the Petitioner to comply with the directive and submit the status report of the same within two months of the issue of this Tariff Order.	Hoppers. - Ash is currently being used to fill low lying areas. - Also, TVNL has written letter to RCD of Jharkhand department to use the ash free of cost. - TVNL has also approached CCL for using ash to fill their abandoned mines.
Details of Loan Repayment Details of Outstanding dues with JSEB	The Petitioner has submitted that due to non-recovery of energy dues from JSEB, repayment of loan has not been undertaken. The amount allowed by way of depreciation is piled up with JSEB as energy dues and the repayment of loan is done only after recovery of dues from JSEB. The Petitioner submitted that it had already provided in the earlier paragraphs of the Petition that as on date 1st Dec 2014 JSEB owes around Rs. 3076 Cr as outstanding dues.	The Commission views that the Petitioner should not link the Repayment of Loan with the recovery of dues from JSEB successor companies and work out a mechanism to liquidate the Govt. Loan either through repayment or conversion of equity.	The Petitioner submits to the Hon'ble Commission that TVNL management had submitted a letter on 7th Jan 2017 to the Government of Jharkhand with 2 proposals regarding the financial restructuring of TVNL's borrowings and its outstanding dues with JUVNL. The details of the said proposals are provided below: - TVNL proposed Government of Jharkhand to convert principal loan amount of INR 665.89 Cr and interest up to INR 1334.01 Cr totalling to INR 1999.90 Cr into equity share capital. Also remaining portion of accrued interest of INR 574.63 Cr along with penal interest of INR 271.11 Cr may be waived off /adjusted with JUVNL. - GoJ may help in realization of outstanding dues with JSEB (INR 3082 Cr) and in turn TVNL shall repay the loan amount of GoJ in full i.e. INR 2845



Directives	Compliance to earlier Directives	Views of the Commission	Current Situation
			Cr. Further GoJ may infuse INR 2845 Cr or at least INR 2000 Cr in equity to TVNL.

Table 9: Status on new directives issued by Hon'ble Commission

S. No.	Directive	Status
1	Minimization of Outages and Strengthening of evacuation arrangement for TTPS	The petitioner is making efforts to reduce the instances of non-evacuation of power on account of outages of transmission lines. The operation of transmission line at 400kV level from present 220kV level will be started post completion of upgradation work in June 2017. The construction of 220 kV TVNL- Jainamore (Bokaro)- Govindpur transmission line being constructed by JUSNL is expected to be completed by December, 2017.
2	Adjustment of Bills & payments/receipt as per Revised Cost of Power Sold to JUVNL/ JBVNL	The petitioner has started refunding/ adjusting the approved Revenue gap for FY 2012-13 and FY 2013-14 in 6 equal monthly instalments from December, 2016 onwards till May, 2017.
3	Filing of True Up Petition for FY 2014-15 and APR for FY 2015-16	The Petitioner submits that the Annual Accounts for FY 2014-15 and FY 2015-16 are yet to be audited and the True up petition for FY 15 and FY 16 shall be filed subsequent to the audit of annual accounts.
4	Data adequacy in next Tariff petition and timelines	The petitioner has prepared the present petition considering the Hon'ble Commission's directive to remove data deficiencies.



CHAPTER 4. CAPITAL EXPENDITURE PLAN

4.1 Capital expenditure Plan

4.1.1 This section deals with actual capital expenditure carried out during FY 2015-16 along with projections for MYT control period for FY 2016-17 to 2020-21.

4.1.2 Following tables summarizes the capital expenditure projections for MYT control period:

Table 10: Capital Investment Plan for FY 2016-17 to FY 2020-21

Capital Expenditure	FY 17	FY 18	FY 19	FY 20	FY 21
Control & Instrumentation	25.88	12.00	22.15	2.55	0.35
Chemistry (OPN)	0.00	0.00	0.65	0.65	0.00
Civil Works	3.51	4.00	35.50	34.00	12.50
MGR System	0.00	0.00	5.00	5.00	10.00
Electrical Dept.(EWD) Works	35.04	24.24	23.20	12.15	11.35
Turbine Dept. (TMD) Works	33.20	11.84	18.08	27.46	50.66
Boiler Dept. (BMD) Works	2.10	17.27	13.82	9.12	13.32
Others	0.69	0.00	0.00	0.00	0.00
Coal Handling Plant (CHP)	0.00	5.00	10.00	16.20	13.00
Total	100.42	74.35	128.40	107.13	111.18

4.2 Capitalization Plan

4.2.1 TVNL has capitalized expenditure of INR 131.40 Cr in FY 2015-16 and as seen from the table above; TVNL projects capitalization during FY 2016-17 as INR 140.10 Cr., INR 80.92 Cr. in FY 2017-18, INR 123.65 Cr, in FY 2018-19, INR 108.78 Cr. in FY 2019-20, INR 110.94 Cr. in FY 2020-21.

4.2.2 TVNL proposes to fund the given capitalization with 70:30 debt-equity ratio with equity infusion expected from internal accruals and/or external equity support and loans from Govt. of Jharkhand. TVNL expects to capitalize the work in progress and capex additions during the year.

4.2.3 TVNL request the Hon'ble Commission to approve the above capital expenditure as well as capitalization for the Second MYT Control Period i.e. FY 2016-17 to FY 2020-21.

Table 11: Capitalisation and proposed funding during FY 2016-17 to FY 2020-21

Particulars	FY 17	FY 18	FY 19	FY 20	FY 21
Opening Capitalization	55.24	15.57	8.99	13.74	12.09



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Particulars	FY 17	FY 18	FY 19	FY 20	FY 21
Capex added during the year	100.42	74.35	128.40	107.13	111.18
Capex Converted to GFA	140.10	80.92	123.65	108.78	110.94
Closing Capitalization	15.57	8.99	13.74	12.09	12.33
Opening GFA	1,588.50	1,728.59	1,809.52	1,933.17	2,041.95
GFA Conversion from Capitalization	140.10	80.92	123.65	108.78	110.94
Closing GFA	1,728.59	1,809.52	1,933.17	2,041.95	2,152.89
Debt (70%)	98.07	56.65	86.56	76.15	77.66
Equity (30%)	42.03	24.28	37.10	32.63	33.28



CHAPTER 5. MULTI YEAR TARIFF PROJECTIONS FOR FY 17 TO FY 21

This section outlines details of fixed and variable cost components proposed by the TVNL under this MYT petition for the control period of FY 2016-17 to 2020-21.

For the purpose of MYT petition all the expense heads has been categorized into Controllable and Uncontrollable factors as shown in the following sections.

5.1 Categorization of Heads of Expenses

5.1.1 For above purpose, TVNL has categorized the expenditures used for the purpose of the MYT petition into Uncontrollable and Controllable expenses:

Table 12: Controllable and Uncontrollable expenses

S. No.	Particular	Category	Remarks
1.	Fuel Expenses	Uncontrollable	
2.	Capital Expenditure	Controllable	Uncontrollable to extent of delay in funding and precarious financial situation of TVNL due to non-payment from JSEB's part.
3.	Depreciation	Controllable	
4.	Operations and Maintenance Expenses	Controllable	
5.	Interest on Loan & Finance Charges	Controllable	
6.	Interest on Working Capital	Controllable	
7.	Return on Equity	Controllable	As per JSERC Regulations.
8.	Taxes on Income	Uncontrollable	

5.2 Details of proposed operational parameters & Variable Cost for FY 2016-17 to FY 2020-21

5.2.1 The proposed variable cost for TVNL for control period of FY 2016-17 to FY 2020-21 are based on the projected operational parameters, GCV's and cost of different fuels for the given control period. The Petitioner in the following section projects the performance of the plant.



5.2.2 This section deals with the approach adopted by petitioner for projecting the operational parameters & Variable Cost for control period of FY 2016-17 to FY 2020-21 based on the provisional accounts for the FY 2015-16.

5.3 Operational Parameters

5.3.1 Following table summarizes the Operational Parameter's projections for MYT control period for FY 2016-17 to FY 2020-21:

Table 13: Operating parameters projected for Second Control Period

Particulars	FY 17	FY 18	FY 19	FY 20	FY 21
PAF (%)	85%	85%	85%	85%	85%
PLF (%)	85%	85%	85%	85%	85%
Station Heat Rate (Kcal/kWh)	3,043	2,908	2,773	2,638	2,503
Aux Cons (%)	9.50%	9.50%	9.50%	9.50%	9.50%
Secondary Oil Cons (ml/kWh)	1.00	1.00	1.00	1.00	1.00
Transit Loss (%)	0.20%	0.20%	0.20%	0.20%	0.20%

5.3.2 The detailed description of the same is provided below:

(a) Plant Availability

Currently, the Plant is operating at the (PAF of 87.36%) PLF of 71.46% for FY 2015-16. However, for the Control Period (FY 2016-17 to FY 2020-21), the Petitioner projects to meet the Normative Annual Plant Availability Factor of 85% as per the clause 8.4 of the JSERC Generation Tariff Regulations, 2015.

(b) Station Heat Rate (SHR)

Petitioner in its endeavour of continuously improving the heat rate of its station strives to improve it further. In FY 2015-16, the Petitioner maintained the Station Heat Rate of 2,753.24 Kcal/kWh. However, for the Control Period (FY 2016-17 to FY 2020-21), the Petitioner projects the Station Heat Rate from 3,043 Kcal/kWh to 2,503 Kcal/kWh in accordance with the clause 8.4 of the JSERC Generation Tariff Regulations, 2015. The year wise projected Station Heat Rate is tabulated below as:



Table 14: Station Heat Rate projected for Second Control Period

Particulars	FY 17	FY 18	FY 19	FY 20	FY 21
Station Heat Rate (kCal/kWh)	3043	2908	2773	2638	2503

The Petitioner requests the Hon'ble Commission to approve the Station Heat Rate as per the aforesaid norms and not as per the actual past performance. The Petitioner proposed that any gains owing to improvement in SHR w.r.t the norms may be allowed to be retained by TVNL and the same could be utilized by the Petitioner for further improvement in the efficiency.

(c) Auxiliary Consumption

In FY 2015-16, the Auxiliary Consumption of the Plant was 11.68%. The petitioner aims & projects to attain auxiliary consumption of 9.50% for the control period FY 2016-17 to FY 2020-21 as per the clause 8.4 of the JSERC Generation Tariff Regulations, 2015.

(d) Secondary Fuel Oil Consumption

In the new regulations, Hon'ble Commission has specified normative secondary fuel oil consumption of 1 ml/kWh for the entire control period. The petitioner would like to submit that average specific oil consumption of TVNL for the past years has been in the range of 2.05-1.15 ml/kWh. In FY 2015-16, the Specific Oil Consumption was 1.153 ml/kWh. Petitioner in its endeavour to improve the performance aims to achieve specific oil consumption of 1 ml/kWh for the control FY 2016-17 to FY 2020-21 as per the clause 8.4 of the JSERC Generation Tariff Regulations, 2015 and requests the Hon'ble Commission to approve the same.

(e) Transit Loss

Petitioner projects to meet the 0.20% as Transit Loss for the Second MYT Control Period. Thus, TVNL requests the Hon'ble JSERC to approve the Transit Loss of 0.20% which is in line with the clause 8.21 of JSERC Generation Tariff Regulations, 2015.

5.4 Variable cost (Energy Charge) computation

5.4.1 As per the clause 8.3 of JSERC Generation Tariff Regulations, 2015,

"Energy charges shall be derived on the basis of the landed fuel cost (LFC) of a generating station (excluding hydro) and shall consist of the following cost:

a) Landed Fuel Cost of primary fuel; and

b) Cost of secondary fuel oil consumption."



5.4.2 Accordingly, various assumptions and computation for determination of Energy Charges is given below:

(a) **Gross Calorific Value of Coal**

As per the Clause 8.24 of JSERC Generation Tariff Regulations, 2015,

“Initially the Base value of gross calorific value of fuel oils and gross calorific value of coal incurred by the Generating Company/ generating Station shall be taken based on actuals of the weighted average gross calorific value of the three preceding months”

Thus, in line with the aforesaid regulations, the Petitioner has considered the Gross Calorific Value of Coal receipt for the preceding three months from the start of Control Period (i.e. Jan’16 to March’16) and based on the same has computed the weighted average GCV of Coal, the computation of which is provided below:

Table 15: Weighted average GCV of coal

S. No.	Particulars	Coal Consumption (MT)	GCV of Coal (kCal/kg)
1	Jan-16	184,129.90	3,622.99
2	Feb-16	158,967.70	3,601.09
3	Mar-16	183,923.20	3,619.73
	Weighted Average GCV of Coal		3,615.25

Thus, as given above, the Petitioner has considered the GCV of Coal as 3,615.25 kCal/kg for the Control Period (FY 2016-17 to FY 2020-21) and requests the Hon’ble Commission to approve the same.

(b) **Cost of Coal**

As per the Clause 8.21 of JSERC Generation Tariff Regulations, 2015,

“The landed fuel cost of primary fuel and secondary fuel for tariff determination shall be based on actual weighted average cost of primary fuel and secondary fuel of the three preceding months and in the absence of landed costs for the three preceding months, latest procurement price of primary fuel and secondary fuel for the generating station, before the start of the tariff period for existing stations and immediately preceding three months in case of new generating stations shall be taken into account.”

Thus, in line with the aforesaid regulations, the Petitioner has considered the Cost of Coal receipt for the preceding three months from the start of Control



Period (i.e. Jan'16 to March'16) and based on the same has computed the weighted average Rate of Coal, the computation of which is provided below:

Table 16: Weighted average rate of coal

S. No.	Particulars	Coal Consumption (MT)	Rate of Coal (INR/MT)
1	Jan-16	184,129.90	2,393.94
2	Feb-16	158,967.70	2,562.67
3	Mar-16	183,923.20	2,431.98
	Weighted Average Rate of Coal		2,458.11

Thus, as given above, the Petitioner has considered the Rate of Coal as INR 2,458.11/MT for the Control Period (FY 2016-17 to FY 2020-21) and requests the Hon'ble Commission to approve the same.

Clause 7.37 of JSERC Generation Tariff Regulations, 2015 states that,

"The cost of fuel in cases covered under clause 7.34 and 7.35 of these Regulations shall be based on the landed cost incurred (taking into account normative transit and handling losses) by the Generating Company and gross calorific value of the fuel as per actual for the three months preceding the first month for which tariff is to be determined and no fuel price escalation shall be provided during the tariff period."

Also, as per the Clause 8.23 of JSERC Generation Tariff Regulations, 2015,

"Any variation in fuel prices on account of change in the Gross Calorific Value (GCV) of coal or gas or liquid fuel shall be adjusted on a monthly basis on the basis of weighted average GCV of coal or gas or liquid fuel in stock, received and burnt and weighted average landed cost incurred by the generating company for procurement of coal, oil, or gas or liquid fuel, as the case may be for a power station."

In view of the aforesaid clauses, the Petitioner has not considered any escalation in Rate of Coal during the Control Period as any escalation in the same may be recovered through Fuel Price Adjustment mechanism for taking into account the escalation in fuel price. This is also in line with the MYT Order dated 30th May 2012 issued by the Hon'ble Commission for previous Control Period.

(c) Gross Calorific Value of Secondary Oil

As per the Clause 8.24 of JSERC Generation Tariff Regulations, 2015,



“Initially the Base value of gross calorific value of fuel oils and gross calorific value of coal incurred by the Generating Company/ generating Station shall be taken based on actuals of the weighted average gross calorific value of the three preceding months”

Thus, in line with the aforesaid regulations, the Petitioner has considered the Gross Calorific Value of Secondary Oil receipt for the preceding three months from the start of Control Period (i.e. Jan'16 to March'16) and based on the same has computed the weighted average GCV of Secondary Oil, the computation of which is provided below:

Table 17: Weighted average GCV of Secondary Oil

S. No.	Particulars	Oil Consumption (KL)	GCV of Oil (kCal/Litre)
1	Jan-16	196.10	9,359.00
2	Feb-16	460.30	9,359.00
3	Mar-16	48.90	9,359.00
	Weighted Average GCV of Secondary Oil		9,359.00

Thus, as given above, the Petitioner has considered the GCV of Oil as 9,359 kCal/Litre for the Control Period (FY 2016-17 to FY 2021-21) and requests the Hon'ble Commission to approve the same.

(d) Cost of Secondary Oil

As per the Clause 8.21 of JSERC Generation Tariff Regulations, 2015,

“The landed fuel cost of primary fuel and secondary fuel for tariff determination shall be based on actual weighted average cost of primary fuel and secondary fuel of the three preceding months,”

Thus, in line with the aforesaid regulations, the Petitioner has considered the Cost of Secondary Oil for the preceding three months from the start of Control Period (i.e. Jan'16 to March'16) and based on the same has computed the weighted average Rate of Secondary Oil, the computation of which is provided below:

Table 18: Weighted average rate of Secondary Oil

S. No.	Particulars	Oil Consumption (KL)	Rate of Oil (INR/KL)
1	Jan-16	196.10	39,475.09
2	Feb-16	460.30	34,897.78
3	Mar-16	48.90	32,508.21



S. No.	Particulars	Oil Consumption (KL)	Rate of Oil (INR/KL)
	Weighted Average Rate of Secondary Oil		36,004.77

Thus, as given above, the Petitioner has considered the Rate of Secondary Oil as INR 36,004.77/KL for the Control Period (FY 2016-17 to FY 2020-21) and requests the Hon'ble Commission to approve the same.

Clause 7.37 of JSERC Generation Tariff Regulations, 2015 states that,

"The cost of fuel in cases covered under clause 7.34 and 7.35 of these Regulations shall be based on the landed cost incurred (taking into account normative transit and handling losses) by the Generating Company and gross calorific value of the fuel as per actual for the three months preceding the first month for which tariff is to be determined and no fuel price escalation shall be provided during the tariff period."

Also, as per the Clause 8.23 of JSERC Generation Tariff Regulations, 2015,

"Any variation in fuel prices on account of change in the Gross Calorific Value (GCV) of coal or gas or liquid fuel shall be adjusted on a monthly basis on the basis of weighted average GCV of coal or gas or liquid fuel in stock, received and burnt and weighted average landed cost incurred by the generating company for procurement of coal, oil, or gas or liquid fuel, as the case may be for a power station."

In view of the aforesaid clauses, the Petitioner has not considered any escalation in Rate of Secondary Oil during the Control Period as any escalation in the same may be recovered through Fuel Price Adjustment mechanism by taking into account the escalation in fuel price. This is also in line with the MYT Order dated 30th May 2012 issued by the Hon'ble Commission for previous Control Period.



(e) **Energy Charge Rate Computation**

The Energy Charge Rate has been computed as per the Clause 8.19 of JSERC Generation Tariff Regulations 2015 which states that,

“8.19 Energy charge rate (ECR) in Rupees per kWh on ex-power plant basis shall be determined to three decimal places in accordance with the following formulae

(a) For coal based stations

$$ECR = \{(GHR - SFC \times CVSF) \times LPPF/CVPF + SFC \times LPSFi + LC \times LPL\} \times 100 / (100 - AUX)\}$$

.....

.....

Where,

AUX - Normative auxiliary energy consumption in percentage

CVPF – (a) Weighted Average Gross calorific value of coal as received, in kCal per kg for coal based stations

(b) Weighted Average Gross calorific value of primary fuel as received in kCal per kg, per litre or per standard cubic meter, as applicable for lignite, gas and liquid fuel based stations.

(c) In case of blending of fuel from different sources, the weighted average Gross calorific value of primary fuel shall be arrived in proportion to blending ratio.

CVSF - Calorific value of secondary fuel, in kCal per ml

ECR - Energy charge rate, in Rupees per kWh sent out.

GHR - Gross station heat rate, in kCal per kWh.

LC = Normative limestone consumption in kg per kWh

LPPF - Weighted average landed price of primary fuel, in Rupees per kg, per litre or per standard cubic metre, as applicable, during the month. (In case of blending of fuel from different sources, the weighted average landed price of primary fuel shall be arrived in proportion to blending ratio)

LPSFi=Weighted Average Landed Price of Secondary Fuel in Rs./ml during the month

LPL = Weighted average landed price of limestone in Rupees per kg.

SFC - Specific fuel oil consumption, in ml per kWh”



On the basis of the aforementioned regulations along with the discussed performance parameters, the Energy Charge Rate MYT control period (FY 17 to FY 21) has been computed and the same is provided in the table given below:

Table 19: Energy Charge computation for Second Control Period

Computation of Energy Charges	Unit	FY 17	FY 18	FY 19	FY 20	FY 21
Gross Generation	MU	3,127	3,127	3,127	3,127	3,127
Net Generation	MU	2,830	2,830	2,830	2,830	2,830
Station Heat Rate	Kcal/kWh	3,043	2,908	2,773	2,638	2,503
Secondary Fuel Oil Consumption	ml/kWh	1	1	1	1	1
Gross Calorific Value of Coal	Kcal/Kg	3,615	3,615	3,615	3,615	3,615
Gross Calorific Value of Oil	Kcal/Kl	9,359	9,359	9,359	9,359	9,359
Total Heat Required	Kcal	9,516,435	9,094,247	8,672,058	8,249,870	7,827,682
Oil Consumption	KL	3,127	3,127	3,127	3,127	3,127
Heat From Oil	Kcal	29,269	29,269	29,269	29,269	29,269
Heat From Coal	Kcal	9,487,166	9,064,978	8,642,790	8,220,602	7,798,413
Transit Loss	%	0.20%	0.20%	0.20%	0.20%	0.20%
Total Coal Consumption including Transit Loss	MT/ year	2,629,469	2,512,455	2,395,441	2,278,427	2,161,413
Cost of Oil	Rs /KL	36,005	36,005	36,005	36,005	36,005
Cost of Coal	Rs /MT	2,458	2,458	2,458	2,458	2,458
Total Cost of Oil	Rs Cr	11.26	11.26	11.26	11.26	11.26
Total Cost of Coal	Rs Cr	646.35	617.59	588.83	560.06	531.30
Energy Charges (Rs/kWh)	Rs/kWh	2.32	2.22	2.12	2.02	1.92

5.5 Details of Fixed Cost for Control Period of FY 2016-17 to 2020-21

5.5.1 This section deals with the approach adopted by petitioner for projecting the fixed cost for control period of FY 2016-17 to 2020-21 based on the finalised accounts for the FY 2015-16. The Revenue Requirement for the control period of FY 2016-17 to FY 2020-21 is projected based on methodologies discussed in detail in subsequent paragraphs.

5.5.2 As per the clause 8.3 of JSERC Generation Tariff Regulations, 2015, the annual fixed charges of a thermal generating station shall consist of the following components:

- (i). Return on Equity;



- (ii). Interest and Financing Charges on Loan Capital;
- (iii). Depreciation;
- (iv). Operation and Maintenance Expenses;
- (v). Interest Charges on Working Capital;
- (vi). Special allowance in lieu of R&M or separate compensation allowance, wherever applicable;

5.5.3 The details of aforesaid ARR components are provided in the subsequent sections.

5.6 Return on Equity

5.6.1 In the previous MYT Tariff Order dated 30th May 2012 for the Control Period FY 2012-13 to FY 2015-16, the Hon'ble Commission had approved the Opening Equity of INR 169.18 Cr. for FY 13 (As on 01st April 2012). For arriving the Opening Equity for FY 2016-17 (01st April 2016), the Petition has added the equity component of Actual Capitalized Assets for FY 13 to FY 16 and the same has been provided below:

Table 20: Actual Capitalization during the period FY 13 to FY 16

Return on Equity	Unit	FY 13	FY 14	FY 15	FY 16
Asset Capitalization (INR Cr.)	Rs Cr	0.97	0.83	2.98	131.40
Equity Component (%)	Rs Cr	30%	30%	30%	100%
Equity Addition (INR Cr.)	Rs Cr	0.29	0.25	0.89	131.40

5.6.2 The Petitioner here would like to submit that the Equity Component of the Asset Capitalized (INR 131.40 Cr.) for FY 16 has been considered as 100% as the same has been completely funded by the cash generated from the core business which is a deviation from the JSERC Generation Tariff Regulations, 2015 and requests the Hon'ble Commission to approve the same.

5.6.3 The Petitioner would further like to submit to the Hon'ble Commission that currently the Shareholder Capital (Equity) of the Petitioner is only INR 105 Cr. whereas as per the Regulation 7 of the JSERC Generation Tariff Regulations, 2015, the Opening Equity for FY 2016-17 should be around INR 475 Cr. (30% of Capitalized Assets of INR 1,588.50 Cr.) which results in huge loss to the utility on account of non-approval of Return on Equity for the difference amount. Thus, in view of the same, the Petitioner requests the Hon'ble Commission to allow 100% equity on the Capitalized Assets for FY 2015-16. Even after considering the 100% equity, the overall opening equity component for FY 2016-17 would become INR 302.02 Cr. which would still be 19% of the total capitalized assets (Less than 30%).



- 5.6.4 Based on the above, the opening equity for the Second MYT Control Period comes out to be INR 302.02 Cr. and the computation of the same is provided below:

Table 21: Computation of Opening Equity for the Control Period

Return on Equity	Unit	FY 13	FY 14	FY 15	FY 16
Equity (Opening Balance)	Rs Cr	169.18	169.47	169.72	170.61
Net additions during the year	Rs Cr	0.29	0.25	0.89	131.40
Equity (Closing Balance)	Rs Cr	169.47	169.72	170.61	302.02

- 5.6.5 The Normative Equity addition during each year of the Control Period has been considered as 30% of the assets capitalized during the year.
- 5.6.6 As per the Clause 7.16 of JSERC Generation Tariff Regulations 2015, the Return on Equity shall be computed on pre-tax basis at the base rate of 15.50%. Since there is no income tax envisaged to be paid by the Petitioner during the Control Period, the Post Tax Rate of Equity has been considered.
- 5.6.7 Based on the opening Normative Equity, Normative Equity addition during the Control Period on account of Additional Capitalization and RoE, the Return on Equity during the Second Control Period (FY 2016-17 to FY 2020-21) is provided in the table given below:

Table 22: Computation of Return on Equity for Second Control Period

Return on Equity	FY 17	FY 18	FY 19	FY 20	FY 21
Equity (Opening Balance)	302.02	344.04	368.32	405.42	438.05
Net additions during the year	42.03	24.28	37.10	32.63	33.28
Equity (Closing Balance)	344.04	368.32	405.42	438.05	471.33
Average Equity	323.03	356.18	386.87	421.73	454.69
Rate of Return on Equity	15.50%	15.50%	15.50%	15.50%	15.50%
Return on Equity	50.07	55.21	59.96	65.37	70.48

- 5.6.8 Thus, the petitioner requests the Hon'ble Commission to approve the Return on Equity as projected above.



5.7 Interest and Financing Charges on Debt

5.7.1 For computation of Interest and Finance Charges, the Petitioner has considered the opening loan balance and interest rates as per the annual accounts of FY 2015-16 and the normative debt addition during the Control Period on account of additional capitalization (as discussed in the previous sections).

5.7.2 In the Current Petition, the Petitioner has considered the Interest and Finance Charges for the control Period as given below:

- (a) **Interest Charges on the loans outstanding as on 31st March 2016** - Refers to the interest charge payable by the Petitioner on the outstanding loans at the end of FY 15. The said loans have been raised from Bihar Government, Jharkhand Government, etc.
- (b) **Interest on additional capitalization** - Interest charges payable for the loans projected to be raised for the additional capitalization projected for the Second Control Period i.e. FY 2016-17 to FY 2020-21. In accordance with the JSERC Generation Tariff Regulations 2015, the Normative Debt on account of Additional Capitalization has been considered as 70%.
- (c) **Repayment of Loan** - The repayment has been considered equal to the depreciation computed for the Control Period in accordance with the clause 7.21 of JSERC Generation Tariff Regulations 2015.
- (d) **Rate of Interest** - The Interest charges for the on-going loans have been considered as per the actual rate of Interest. For the new loans projected during the Control Period, the Interest Rate has been considered in accordance with clause 7.23 of JSERC Generation Tariff Regulations, 2015, i.e. SBI Base Rate plus 200 basis points. As on 01st April 2016, the SBI Base Rate was 9.30% and accordingly, the Interest Rate on Normative Loan comes out to be 11.30%.

5.7.3 Based on the Opening Actual Debt, Normative Debt addition during the Control Period on account of Additional Capitalization and Rate of Interest, the Interest and Finance Charges on Debt during the Second Control Period (FY 2016-17 to FY 2020-21) is provided in the table given below and requests the Hon'ble Commission to approve the same:

Table 23: Debt Schedule for the Second Control Period

Interest on Loan	FY 17	FY 18	FY 19	FY 20	FY 21
Opening Balance	3016.09	3031.37	2999.40	2991.93	2967.92
Loan received during the Year	98.07	56.65	86.56	76.15	77.66
Repayment during the year	82.78	88.62	94.02	100.16	105.97
Balance O/S at the end of the year	3031.37	2999.40	2991.93	2967.92	2939.61



Table 24: Computation of Interest on Debt for Second Control Period

Revenue Requirement	FY 17	FY 18	FY 19	FY 20	FY 21
Bihar Govt.	78.09	75.90	73.63	71.28	68.85
Jharkhand Govt.	7.31	7.11	6.89	6.67	6.44
Interest payable on Bihar Govt. Loan	51.36	49.93	48.43	46.88	45.28
Interest payable on Jharkhand Govt. Loan	2.45	2.38	2.31	2.23	2.16
Sub Total	139.21	135.31	131.26	127.07	122.73
Other Interest & Finance Charges	0.01	0.01	0.01	0.01	0.01
Interest on new loan for Additional Capitalization	5.47	13.93	21.48	29.83	37.35
Total	144.68	149.25	152.75	156.91	160.09

5.8 Depreciation

- 5.8.1 Based on the Opening GFA and asset addition during each year of the Control Period, Depreciation has been computed. The depreciation rates have been computed as per JSERC Generation Tariff Regulations, 2015.
- 5.8.2 Asset-wise details have been annexed in the Format-7 of the Tariff filing formats as Annexure 2.
- 5.8.3 The depreciation projected for the Control Period is provided in the table given below:

Table 25: Projected Depreciation for Second Control Period

Depreciation	FY 17	FY 18	FY 19	FY 20	FY 21
Depreciation for Plant	82.49	88.31	93.69	99.81	105.60
Depreciation for HO	0.30	0.32	0.33	0.35	0.37
Total Depreciation	82.78	88.62	94.02	100.16	105.97



5.9 Operation and Maintenance Expenses

5.9.1 The O&M expenses include expenditure incurred in the operation and maintenance of the generating station, including employee cost, repairs and maintenance, consumption of stores and spares, water charges, ash disposal, pollution control cess, insurance and other administrative and general expenses of the Petitioner corporate office at Ranchi.

5.9.2 As per the clause 7.40 of JSERC Generation Tariff Regulations, 2015; O&M Expenses Comprises of:

- a) Salaries, wages, pension contribution and other employee costs;
- b) Administrative and General costs;
- c) Repairs and maintenance expenses; and
- d) Other miscellaneous expenses statutory levies and taxes (except corporate income tax)

5.9.3 Following sections summarizes the O&M expenses projections for the MYT control period:

5.9.4 Employee Expenses

As per the clause 7.41 (a) of the JSERC Generation Tariff Regulations, 2015,

“The Applicant shall submit details of O&M expenses as required by the Commission. The O&M expenses excluding terminal liabilities for the Base Year shall be determined based on accounting statements for the base year, estimates of the Generating Company for relevant years and other factors considered relevant.”

The Employee Expenses for the base year i.e. FY 2015-16 excluding the Terminal Benefits is provided below:

Table 26: Employee Expense for FY 2015-16

Employee Expenses	FY 2015-16
Salaries	36.57
Honorarium/Overtime	1.39
Pay Revision Arrears	
Sub Total	37.96
Medical Expenses Reimbursement	0.40
Earned Leave Encashment	1.92
Any Other Item	0.01
Payment Under Workman's Compensation And Gratuity	0.32



Employee Expenses	FY 2015-16
Sub Total	2.66
Contribution to Terminal Benefits	21.72
Any Other Item	
Grand Total	62.34

The employee costs (except the Terminal Benefits) for the control period has been evaluated on the basis of per unit employee cost adjusted for an inflationary increase of 5.88% (based on 20% weightage of WPI & 80% weightage of CPI) over FY 2015-16 per unit employee cost.

The Employee strength projection of TVNL during the Control Period is provided below:

Table 27: Employee Strength projection during the Control Period

Employee Pay Bands		FY 17	FY 18	FY 19	FY 20	FY 21
PB 1	37400/- to 67000/ (GP 10000, GP 9000, GP 8700)	16	12	8	1	1
PB 2	15600 to 39100/- (GP 6600)	29	29	29	29	29
PB 3	9300 to 34800/- (GP 5600, GP 5500, GP 5300, GP 4600, GP 4500, GP 4300, GP 4200)	196	215	215	215	215
PB 4	5200 to 20200 (GP 3300, GP 3000, GP 2700, GP 2400, GP 2100)	389	478	478	467	465
	Total	630	734	730	712	710

Also as per the clause 7.41 (b) of the JSERC Generation Tariff Regulations, 2015, Terminal Liabilities will be approved as per actual submitted by the Generating Company or be established through actuarial studies. Thus, the Terminal Benefits as per the accounts of FY 2015-16 has been considered for projection of Terminal Benefits during the Control Period. The same shall be trued up as per the actual terminal liabilities incurred by the Petitioner after due prudence check by the Commission.

In addition to the above, the Petitioner submits that on account of sixth pay Commission, the Petitioner expects the additional liability of INR 3 Cr. for FY 2016-17 and the same has also been considered for projection of Employee Expenses for the Second MYT Control Period.



Based on the above, the Employee Expenses projected for the Control Period (FY 2016-17 to FY 2020-21) is tabulated below:

Table 28: Employee cost for the Control Period

Employee Expenses	FY 17	FY 18	FY 19	FY 20	FY 21
Salaries	40.86	50.40	53.07	54.81	57.86
Honorarium/Overtime	1.56	1.92	2.02	2.09	2.21
Pay Revision Arrears					
Sub Total	42.41	52.32	55.09	56.89	60.07
Medical Expenses Reimbursement	0.45	0.56	0.59	0.60	0.64
Pension	2.14	2.64	2.78	2.87	3.03
Payment Under Workman's Compensation And Gratuity	0.36	0.45	0.47	0.49	0.51
Any Other Item	0.02	0.02	0.02	0.02	0.02
Sub Total	2.97	3.66	3.86	3.98	4.20
Contribution to Terminal Benefits	21.72	21.72	21.72	21.72	21.72
Total	67.11	77.71	80.67	82.60	86.00
Additional Impact on account of Sixth Pay Commission	3.00	4.13	4.35	4.50	4.75
Grand Total	70.11	81.84	85.03	87.10	90.74

5.9.5 A&G Expenses

As per the clause 7.41 (a) of the JSERC Generation Tariff Regulations, 2015,

"The Applicant shall submit details of O&M expenses as required by the Commission. The O&M expenses excluding terminal liabilities for the Base Year shall be determined based on accounting statements for the base year, estimates of the Generating Company for relevant years and other factors considered relevant."

The A&G Expenses for the base year i.e. FY 2015-16 is provided below:

Table 29: Actual A&G Expenses for FY 2015-16

A&G Expenses	FY 16
Lease/Rent	0.58
Insurance	0.32
Telephone, Postage, Telegram & Telex Charges	0.19
Consultancy Charges	0.29
Other Professional Charges	0.11
Conveyance and Travelling	0.26
Security/Service Charges Paid to Outside Agencies	9.07



**FILING OF PETITION FOR
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A&G Expenses	FY 16
Late Penalty/ Intt Charges	0.01
License and Registration Fees	0.07
Rates & Taxes	0.02
Sub Total	10.92
Fee and Subscription of Books and Periodicals	0.67
Printing and Stationary	0.09
Advertisement Expenses (Other than Purchase	0.13
Electricity Charges to Offices	0.19
Water Charges	0.13
Meeting and Entertainment Charges	0.12
Miscellaneous Expenses	0.05
Sub Total	1.39
Legal Charges	0.26
Auditor's Fee	0.05
Vehicle Running Expenses Truck/Delivery Van	0.13
Vehicle Hiring Expenses Truck/Delivery Van	1.08
Others	0.00
Sub Total	1.52
Direction and Supervision Charges	
Grand Total	13.84

The costs of MYT control period for FY 2016-17 to FY 2020-21 have been evaluated on the basis of an inflation rate of 5.88% (based on 20% weightage of WPI & 80% weightage of CPI) for % increase in FY 2015-16 on year-on-year basis.

Based on the above, the projected A&G Expenses for the MYT Control Period (FY 17 to FY 21) are provided below:

Table 30: A&G Expenses projected for Second Control Period

A&G Expenses	FY 17	FY 18	FY 19	FY 20	FY 21
Lease/Rent	0.62	0.65	0.69	0.73	0.77
Insurance	0.34	0.36	0.38	0.41	0.43
Telephone, Postage, Telegram & Telex Charges	0.20	0.21	0.22	0.23	0.25
Consultancy Charges	0.31	0.32	0.34	0.36	0.38
Other Professional Charges	0.12	0.12	0.13	0.14	0.15



**FILING OF PETITION FOR
MULTI YEAR TARIFF FOR CONTROL PERIOD FROM FY 2016-17 TO FY 2020-21**

A&G Expenses	FY 17	FY 18	FY 19	FY 20	FY 21
Conveyance and Travelling	0.27	0.29	0.31	0.33	0.34
Security/Service Charges Paid to Outside Agencies	9.60	10.17	10.76	11.40	12.07
Late Penalty/ Int. Charges	0.01	0.01	0.02	0.02	0.02
License and Registration Fees	0.07	0.08	0.08	0.08	0.09
Rates & Taxes	0.02	0.03	0.03	0.03	0.03
Sub Total	11.56	12.24	12.96	13.73	14.53
Fee and Subscription of Books and Periodicals	0.71	0.75	0.80	0.84	0.89
Printing and Stationary	0.10	0.11	0.11	0.12	0.13
Advertisement Expenses (Other than Purchase	0.14	0.15	0.16	0.17	0.18
Electricity Charges to Offices	0.21	0.22	0.23	0.24	0.26
Water Charges	0.13	0.14	0.15	0.16	0.17
Meeting and Entertainment Charges	0.13	0.14	0.15	0.16	0.16
Miscellaneous Expenses	0.05	0.05	0.05	0.06	0.06
Sub Total	1.47	1.56	1.65	1.75	1.85
Legal Charges	0.28	0.29	0.31	0.33	0.35
Auditor's Fee	0.05	0.05	0.06	0.06	0.06
Vehicle Running Expenses Truck/Delivery Van	0.14	0.15	0.16	0.17	0.18
Vehicle Hiring Expenses Truck/Delivery Van	1.14	1.21	1.28	1.36	1.44
Others	0.00	0.00	0.00	0.00	0.00
Sub Total	1.61	1.71	1.81	1.91	2.03
Direction and Supervision Charges					
Grand Total	14.65	15.51	16.42	17.39	18.41

Therefore, the petitioner requests the Hon'ble Commission to approve the A&G expenses as projected above.



5.9.6 R&M Expenses

As per the clause 7.41 (a) of the JSERC Generation Tariff Regulations, 2015,

“The Applicant shall submit details of O&M expenses as required by the Commission. The O&M expenses excluding terminal liabilities for the Base Year shall be determined based on accounting statements for the base year, estimates of the Generating Company for relevant years and other factors considered relevant.”

The R&M Expenses for the base year i.e. FY 2015-16 is provided below:

Table 31: R&M expenses for FY 2015-16

R&M Expenses	FY 16
Plant and Machinery	65.39
Building	1.26
Civil Works	6.03
Hydraulic Works	0.41
Lines, Cables Net Works etc.	-
Vehicles	0.02
Furniture and Fixtures	-
Office Equipments	0.10
Station Supplies	-
Other Credits to R&M Charges	8.32
Total	81.54
Any other items (Capitalization)	-
Total	81.54

**Capital maintenance portion has been excluded from R&M expenses*

The costs of MYT control period for FY 2016-17 to FY 2020-21 have been evaluated on the basis of an inflation rate of 5.88% (based on 20% weightage of WPI & 80% weightage of CPI) for % increase in FY 2015-16 on year-on-year basis.

Based on the above, the projected R&M Expenses for the MYT Control Period (FY 17 to FY 21) are provided below:

Table 32: R&M Expenses for Second Control Period

R&M Expenses	FY 17	FY 18	FY 19	FY 20	FY 21
Plant and Machinery	69.23	73.30	77.61	82.17	87.00
Building	1.34	1.41	1.50	1.58	1.68
Civil Works	6.39	6.76	7.16	7.58	8.03
Hydraulic Works	0.44	0.46	0.49	0.52	0.55



R&M Expenses	FY 17	FY 18	FY 19	FY 20	FY 21
Lines, Cables Net Works etc.	-	-	-	-	-
Vehicles	0.02	0.03	0.03	0.03	0.03
Furniture and Fixtures	-	-	-	-	-
Office Equipments	0.10	0.11	0.12	0.12	0.13
Station Supplies	-	-	-	-	-
Other Credits to R&M Charges	8.81	9.33	9.88	10.46	11.07
Total	86.33	91.41	96.78	102.47	108.49
Any other items (Capitalization)	-	-	-	-	-
Total	86.33	91.41	96.78	102.47	108.49

Therefore, the petitioner requests the Hon'ble Commission to approve the R&M expenses as projected above.

5.9.7 **O&M Expenses Summary**

Following tables summarizes the O&M expenses Summary for MYT control period (FY 2016-17 to FY 2020-21):

Table 33: Summary of O&M Expenses for Second Control Period

O&M Expenses	FY 17	FY 18	FY 19	FY 20	FY 21
Employee Expenses	70.11	81.84	85.03	87.10	90.74
A&G Expenses	14.65	15.51	16.42	17.39	18.41
R&M Expenses	86.33	91.41	96.78	102.47	108.49
Total	171.09	188.76	198.23	206.96	217.65

Therefore, the Petitioner requests the Hon'ble Commission to approve the O&M expenses as projected above.



5.10 Interest Charges on Working Capital

5.10.1 As per the clause 7.34 of JSERC Generation Tariff Regulations 2015, the Working Capital requirement for coal-based generating stations shall contain the following components:

- a. Cost of coal or lignite and limestone towards stock, if applicable, for 15 days for pit-head generating stations and 30 days for non-pit-head generating stations for generation corresponding to the normative annual plant availability factor or the maximum coal/lignite stock storage capacity whichever is lower;
- b. Cost of coal or lignite and limestone for 30 days for generation corresponding to normative annual plant availability factor;
- c. Cost of secondary fuel oil for two months for generation corresponding to the Normative Annual Plant Availability Factor, and in case of use of more than one secondary fuel oil, cost of fuel oil stock for the main secondary fuel oil;
- d. Maintenance spares @ 20% of Operation and maintenance expenses specified in clauses 7.40 to 7.48 of these regulations;
- e. Operation and Maintenance expenses for 1 month; and
- f. Receivables equivalent to 2 months of capacity charges and energy charges for sale of electricity calculated on the Normative Annual Plant Availability Factor.

5.10.2 Following table summarizes the Working Capital expenses projections for MYT control period (FY 2016-17 – FY 2020-21):

Table 34: Working Capital Requirement for Second Control Period

Interest on Working Capital	Unit	FY 17	FY 18	FY 19	FY 20	FY 21
Cost of Coal for 15 days for stocking	Rs Cr	26.56	25.38	24.20	23.02	21.83
Cost of Coal for 30 days for generation	Rs Cr	53.12	50.76	48.40	46.03	43.67
Cost of Secondary Oil for 2 months	Rs Cr	1.88	1.88	1.88	1.88	1.88
Maintenance Spares @ 20% of O&M	Rs Cr	34.22	37.75	39.65	41.39	43.53
O&M Expenses for 1 month	Rs Cr	14.26	15.73	16.52	17.25	18.14
Receivable equivalent to 2 months average billing	Rs Cr	187.43	188.22	187.24	186.48	185.79
Total Working Capital Requirement	Rs Cr	317.47	319.72	317.88	316.04	314.84



- 5.10.3 As per the clause 7.38, the Interest on Working Capital shall be on normative basis and shall be considered as the bank rate as on 01st April 2016 which is Base Rate as per State Bank of India plus 350 basis points and the same comes out to be 12.80% and the same has been considered for computation of Interest on Working Capital.
- 5.10.4 Based on the above, the Interest on Working Capital computed is tabulated as given below:

Table 35: Interest on Working Capital for Second Control Period

Interest on Working Capital	FY 17	FY 18	FY 19	FY 20	FY 21
Total Working Capital requirement	317.47	319.72	317.88	316.04	314.84
Rate of Interest	12.80%	12.80%	12.80%	12.80%	12.80%
Interest on Working Capital	40.64	40.92	40.69	40.45	40.30

- 5.10.5 Therefore, the Petitioner requests the Hon'ble Commission to approve the above mentioned interest on working capital.

5.11 Tax on Income

- 5.11.1 The norms of the Hon'ble JSERC and all the other SERCs/ CERC allow the Tax on Income earned from the generation business will be reimbursable from the buyer of the power. Thus, TVNL requests the Hon'ble Commission to approve reimbursement of Tax on Income that shall be levied upon TVNL through the generation business as per the Regulations.
- 5.11.2 Petitioner does foresee; pending repayment of existing dues by JUVNL / JBVNL; there would be no assessable income for income tax and hence no tax on income has been estimated for the Control Period (FY 17 to FY 21).
- 5.11.3 At this point, TVNL hasn't estimated the impact of taxes that could become applicable. However, if the TVNL does pays any taxes then, TVNL requests Hon'ble JSERC to approve the same as & when accrued upon.



5.12 Water Charges

- 5.12.1 The Petitioner submits that the Water Resource Department has allocated 15.50 MCM of water for industrial purpose and 0.75 MCM for municipal purpose at the charges of INR 4.50 per 1000 gallons of water (for industrial purpose)
- 5.12.2 Based on the same, the Petitioner has incurred the liability of INR 1 Cr. per month as water charges to water resource department. The Petitioner projects the same amount for the Control Period (FY 17 to FY 21) and requests the Hon'ble Commission to approve the same.
- 5.12.3 However, the Petitioner submits that According to Water Department, the Petitioner has to pay water charges based on the entire water intake by them instead of the quantity of water lost due to evaporation as the water is discharged back into the main water body after using it in the power plant. The said issue was raised by the Water Department around 2008 – 2009 and a Committee was constituted by them in 2011 for the said matter.
- 5.12.4 After discussion with the concerned authorities of Water Resource Department, the Water Department finally agreed in a joint meeting with TVNL that the latter shall be liable to pay water charges only for the quantity of water lost due to evaporation losses and not for the total quantity of water taken in the plant. However no formal communication has been received by TVNL regarding the same from the Water Department and TVNL shall pay their water charges in future as per the final communication by Water Resource Department regarding the aforesaid matter.
- 5.12.5 In view of the above, the Petitioner requests the Hon'ble Commission to approve the water charges liability as incurred by the Petitioner for FY 2015-16. The Petitioner also proposes that the actual charges based on the formal communication by Water Resource department may be tried up at a later stage.

5.13 Summary of Total Fixed Cost for FY 2016-17 to FY 2020-21

- 5.13.1 Based on above discussed section, the table given below summarises total fixed cost projected for each year of the Control Period (FY 2016-17 to 2020-21):
- 5.13.2 The Petitioner requests the Hon'ble Commission to approve the fixed cost as summarized above.

Table 36: Fixed Cost projection for Second Control Period

Fixed Cost Summary	FY 17	FY 18	FY 19	FY 20	FY 21
O&M Expenses	171.09	188.76	198.23	206.96	217.65
Depreciation	82.78	88.62	94.02	100.16	105.97
Interest on Loan	144.68	149.25	152.75	156.91	160.09



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Fixed Cost Summary	FY 17	FY 18	FY 19	FY 20	FY 21
Return on Equity	50.07	55.21	59.96	65.37	70.48
Interest on Working Capital	40.64	40.92	40.69	40.45	40.30
Water Charges	12.00	12.00	12.00	12.00	12.00
Less: Non-Tariff Income	(34.29)	(34.29)	(34.29)	(34.29)	(34.29)
Total fixed Cost	466.97	500.47	523.37	547.56	572.19



CHAPTER 6. REVENUE REQUIREMENT

6.1 Non-Tariff Income

6.1.1 Following table summarizes the Non-Tariff Computations for MYT control period along with the Base Year (FY 2015-16):

Table 37: Non-tariff Income for the Control Period

Non-Tariff Income	FY 17	FY 18	FY 19	FY 20	FY 21
Income from Investment, Fixed & Call Deposits	33.09	33.09	33.09	33.09	33.09
Income/Fee/Collection against staff welfare activities	1.20	1.20	1.20	1.20	1.20
Total	34.29	34.29	34.29	34.29	34.29

6.1.2 The Petitioner has projected the non-tariff income equal to the FY 2015-16 figures in line with the Hon'ble Commission's approach for the first MYT Order and requests the Hon'ble Commission to approve the Non-Tariff income as summarized above.

6.2 Revenue Requirement for Control period

6.2.1 The sum of fixed and energy costs gives the total revenue requirement of the company for the Base Year FY 2015-16 and for MYT control period FY 2016-17 to FY 2020-21. The revenue requirement less other income will be the revenue to be raised through tariff:

6.2.2 Following table summarizes the Revenue requirement Computations for MYT control period (FY 17 to FY 21):

Table 38: Revenue requirement for Second Control Period

Revenue Requirement	Unit	FY 17	FY 18	FY 19	FY 20	FY 21
Fixed Charges	Rs Cr	501.26	534.76	557.66	581.85	606.48
Energy Charge Rate	Rs/kWh	2.32	2.22	2.12	2.02	1.92
Other Income	Rs Cr	(34.29)	(34.29)	(34.29)	(34.29)	(34.29)
Fixed Income to be raised through tariff	Rs Cr	466.97	500.47	523.37	547.56	572.19

6.2.3 The Petitioner requests the Hon'ble Commission to approve the above revenue requirement projection for the MYT control period.



CHAPTER 7. FUEL PRICE ADJUSTMENT

7.1 FPPPA Formula

7.1.1 The Hon'ble Commission has notified the following FPA formula in the JSERC Generation Tariff Regulations, 2015. Clause 8.22 of the same states that:

"The Fuel Price Adjustment (FPA) applicable for calculation of Energy Charges is as follows:

(a) For coal-based generating stations, $FPA = A + B$

Where,

FPA – Fuel price Adjustment for a month in Paise/kWh Sent out;

A – Fuel price adjustment for Secondary Fuel oil in Paise/kWh sent out;

B – Fuel price adjustment for Coal in Paise/kWh sent out;

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Where,

SFCn = Normative Specific Fuel Oil consumption in l/kWh;

SHRn = Normative Gross Station Heat Rate in Kcal/kWh;

ACn = Normative Auxiliary consumption in percentage;

Pom = Weighted average price of fuel oil on as consumed basis during the month in Rs. /KL;

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..... Ks = Base value of Gross Calorific Value of Gas or Liquid fuel as taken determination of base energy charge in tariff order in Kcal/ SCM or kCal/Litre or kCal/Kg;"

Hon'ble Commission is requested to approve Station Heat Rate, Auxiliary Consumption, Transit Loss, Specific Oil Consumption and Plant Availability Factors. TVNL shall claim the fuel cost on the basis of prevailing landed fuel cost, actual GCV received on the basis of above approved parameters as per the New Regulations.



CHAPTER 8. PRAYER

TVNL respectfully prays to the Hon'ble Commission;

- i. To admit this MYT Petition for Control Period for FY 2016-17 to FY 2020-21.
- ii. To approve Revenue Requirement and Capital Expenditure for MYT Control Period for FY 2016-17 to FY 2020-21.
- iii. To approve the operational and financial parameters as proposed by TVNL considering the constraints of the old machines, and consider the same for recovery of full fixed cost.
- iv. To approve the Station's operating parameters viz. PAF, Auxiliary Consumption, Station Heat Rate, Transit Loss, Specific Oil Consumption and actual fuel rate for recovery of variable cost considering the constraints of the old machines as well as site specific constraints.
- v. To condone the delay in filing this Petition.
- vi. To grant any other relief as the Hon'ble Commission may consider appropriate.
- vii. To approve the ARR based on the operating parameters as per the norms provided in the Regulation and allow the Petitioner to retain the gains on account of the efficiency improvement in the Operating Parameters.
- viii. The Petitioner craves leave of the Hon'ble Commission to allow further submissions, addition and alteration to this Petition as may be necessary from time to time.
- ix. Pass any other order as the Hon'ble Commission may deem fit and appropriate under the circumstances of the case and in the interest of justice.

Declaration that the subject matter of the petition has not been raised by the petitioner before any other competent forum, and that no other competent forum is currently seized of the matter or has passed any orders in relation thereto.



Abbreviations

APH	Air Pre-Heater
ARR	Aggregate Revenue Requirement
BFP	Boiler Feed Pump
CAGR	Compounded Annual Growth Rate
CCL	Central Coalfields Limited
CEA	Central Electricity Authority
CERC	Central Electricity Regulatory Corporation
Cr	Crores
Control Period	FY 2016-17 to FY 2020-21
CW	Coolant Water
D/C	Direct Current
DPR	Detailed Project Report
DVC	Damodar Valley Corporation
FD	Forced Draught
FY	Financial Year
GoJ	Government of Jharkhand
ID	Induced Draught
JSEB	Jharkhand State Electricity Board
KV	Kilo Volt
KWh	Kilo Watt Hour
MGR	Merry Go Round
MOEF	Ministry of Environment and Forest
MUs	Million Units
MW	Mega Watt
MYT	Multi Year Tariff
O&M	Operation & Maintenance
PAT	Perform, Achieve and Trade
PAF	Plant Availability Factor
PPA	Power Purchase Agreement
PLF	Plant Load Factor
PLR	Prime Lending Rate
R E	Revised Estimates



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R & M	Renovation & Modernisation/ Repair & Maintenance
ROE	Return on Equity
Rs	Indian National Rupees
S/S	Sub Station
SERC	State Electricity Regulatory Corporation
SHR	Station Heat Rate
TCS	Tata Consultancy Services
SAP	IT Company
TTPS	Tenughat Thermal Power Station
TVNL	Tenughat Vidyut Nigam Limited



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Annexure 1: PROVISIONAL ACCOUNTS FOR FY 2015-16



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Annexure 2: TARIFF FILING FORMATS



Picture 1: Tenughat Vidyut Nigam Limited (TVNL) 2*210 MW Power Plant

.....Powering Jharkhand